Ann Arbor Hands-On Museum
Final Report Out

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April 17th, 2026

Background and Problem Statement

"Nonprofit whose mission is to create moments of discovery that inspire curiosity for STEAM"
-AAHOM

Problem Statement

The Ann Arbor Hands-On Museum would like to **track year over year attendance with more segmentation, and insights into what donors to target** in order to make strategic decisions that bolster their participation and donation growth

Deliverable Overview

Attendance Dashboard

- YoY comparison of 52 weeks attendance
- Creating a graph of projected signups vs actual attendees
- **23** Visualizations developed

Membership Dashboard

- Highlight YoY engagement trends with museum sign ups in person vs. online
- Used to understand museum attendance by category
- **2** Visualizations developed

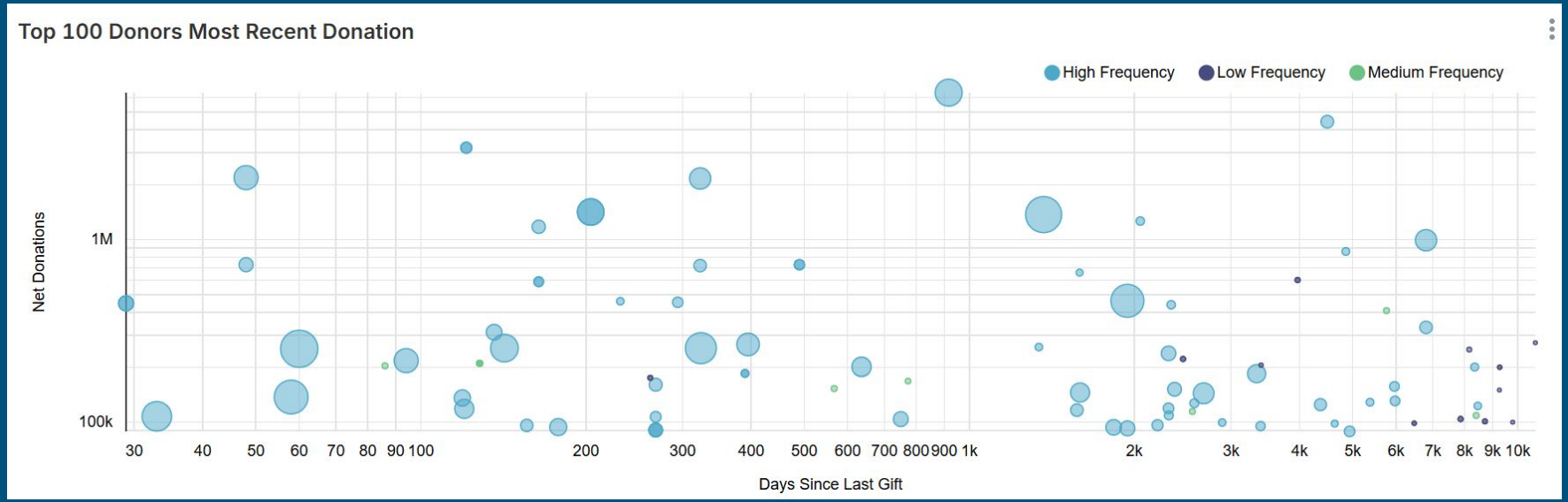
Donor Insight Dashboard

- Used to understand giving patterns, retention, and engagement
- Provide insights that support long-term fundraising growth
- **9** Visualizations developed.

Key Insight:

All dashboards are used to **empower staff** with clear, **data-driven** information to guide smarter decisions and strengthen AAHOM's **programs, operations, and donor engagement**

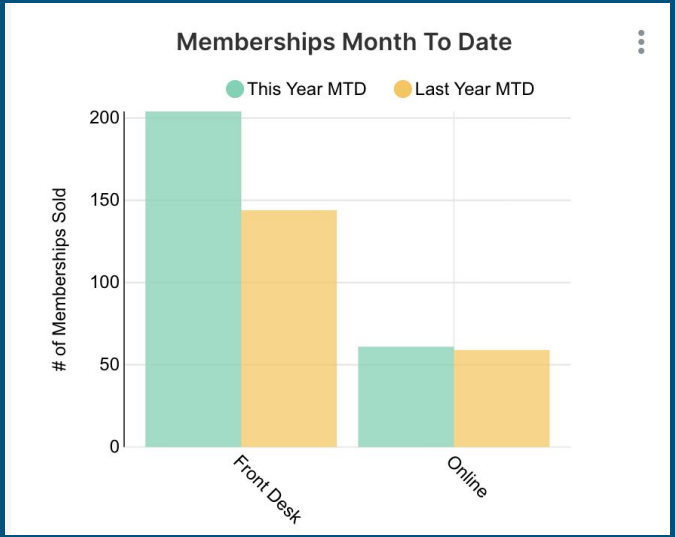
Visualization #1: Donor Recency

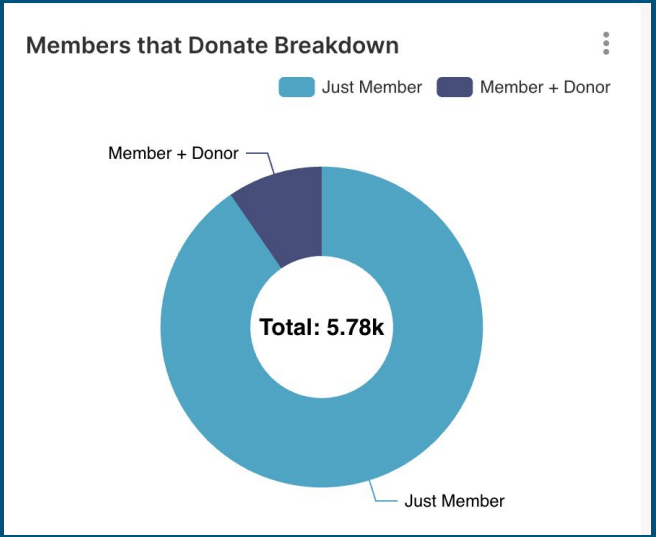


Key Insight:

Overall, the data shows a strong active base and long-term lapse segment that may benefit from **reactivation efforts**. This is important because it helps the museum prioritize outreach and increase donor retention

Visualization #2: Memberships Month to Date & Member/Donor Overlap

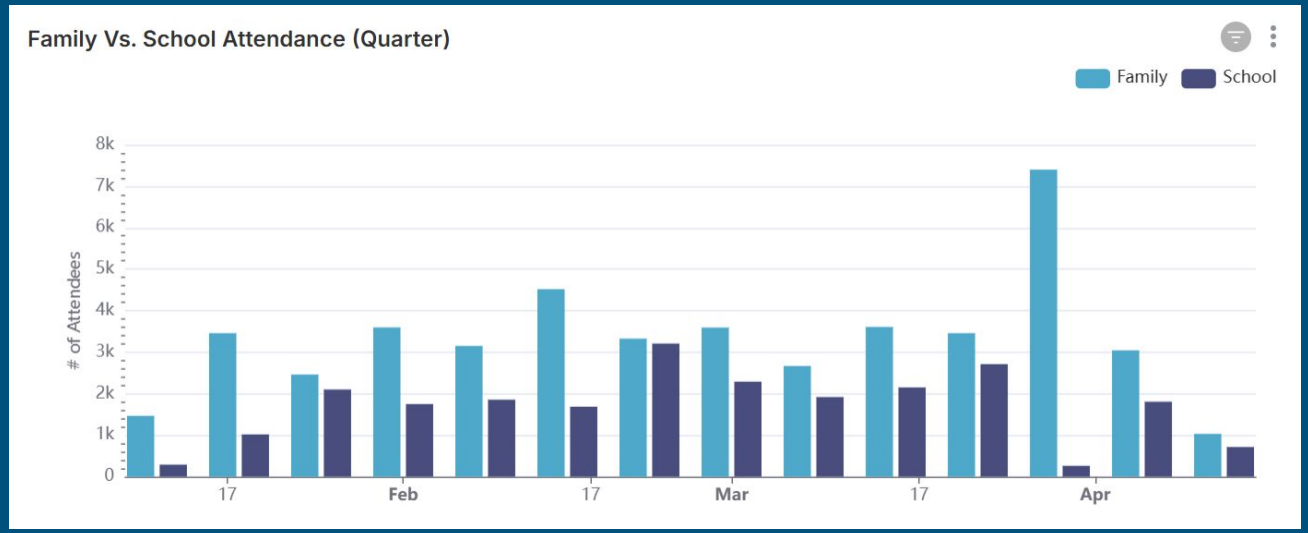




Key Insight:

The data shows that the majority of AAHOM's members purchase in person and do not contribute to the museum outside of membership costs. This presents an opportunity for donor growth in a key demographic

Visualization #3: Family vs School Attendance



Context

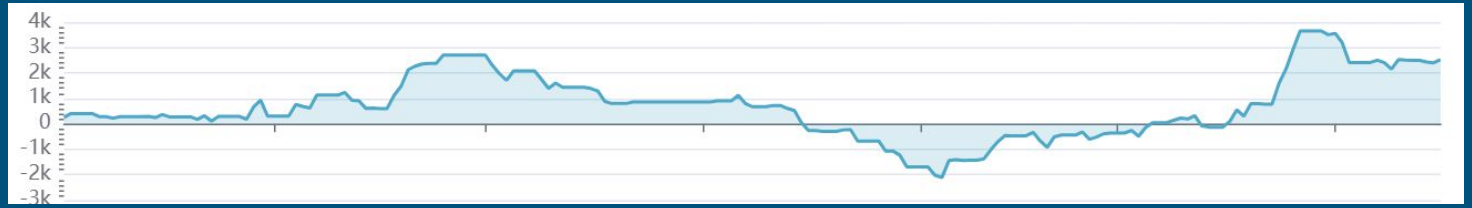
Family events include: Birthday parties, rentals, camps, and public programs

School events include: Field trips, distance learning, field trips at LSNC, outreach programs, Yankee Air Museum, and Trinity HES

Key Insight:

To **maximize overall attendance**, the museum should prioritize **high-yield family programming**, replicate the strategies behind the **late-March attendance spike**, and **restructure** its school outreach to boost consistently lower engagement levels

Visualization #4: Cumulative Field Trip Attendance



Key Insight:

This chart compares current field trip (cumulative) attendance accompanied by a visual to magnify the difference YoY to **identify seasonal trends and growth**, showing a volatile, but general **upward trajectory** in school group visits as we move into May 2026

Handoff Process

April 17th, 2026

Have all dashboards completed with implemented client feedback

April 20rd, 2026

Finalize documentation, present at Expo!

April 23rd, 2026

Train staff on dashboard usage and deliver data refresh procedures on long term use

Challenges

Navigating analysis across multiple platforms

- We had to adjust SQL queries to fix incorrect donor counts or create new chart.
- Difficult to maintain consistency, keep analysis standard and organized

Mitigation Strategy

Use Gantt chart as a centralized tracker to organize tasks, monitor progress and ensure alignment on timelines / responsibilities

Dashboard Performance & Latency

- Superset analysis across multiple dashboards causes massive amounts of historical data every time a user opens or filters the dashboard.
- The dashboard taking too long can delay efforts for AAHOM employees

Mitigation Strategy

Instead of running the heavy query "live," set up a scheduled process to run it once overnight and save the results into a simple, table. By doing this, the dashboard will load instantly with fresh data

Project Documentation

1. **User Documentation**
   - Explain and navigates dashboards in Apache Superset
   - Understand Attendance, Membership and donor insights dashboards objectifs
   - Explore visualizations, understand their description, and purpose
   - Best practices for daily decision making
2. **Data Refresh & Troubleshooting**
   - Developed a technical guide for maintaining dashboards (refresh frequency)
   - How to solve common issues (missing data, slow dashboards, errors)
3. **Training and Handoff**
   - Conducted a 30-minute training session with museum staff

Ann Arbor HANDS-ON Museum

Thank you!